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DRAFT (SAMPLE DOC)

Work Order Inbound REST APIs

The Work Order Inbound REST APIs receive work orders from an external application which are then processed by Gridscale X Meter Data Management (formerly known as EnergyIP MDM). The external system sends work orders as Device Control Transaction (DCT) requests. For example, the external system may raise a remote disconnect request for a customer whose account is past due. The Work Order Inbound REST APIs are used to create work orders, reschedule work orders, check work order statuses, cancel work orders, and send work status notifications to the MDM. These APIs are part of the `workorder-inbound-api` application. To run the Work Order Inbound REST APIs, you must configure the `workorder-inbound-api` application, see [Configuring the Work Order Inbound API Application for REST Interface](#) for details.

- [Supported Operations](#)
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Supported Operations

The following table shows the supported operations for inbound work orders.

Function	Operation	Method
Create a work order for a specified DCT operation	<code>/workorder-inbound-api/v1/work/create</code>	POST
Reschedule an inbound work order for a specified DCT operation	<code>/workorder-inbound-api/v1/work/reschedule</code>	POST
Fetch the status of an inbound work order for a specified DCT operation	<code>/workorder-inbound-api/v1/work/status</code>	POST
Cancel an inbound work order for a specified DCT operation	<code>/workorder-inbound-api/v1/work/cancel</code>	POST
Send a work status notification to MDM when a field investigation is done or when updates are made to the field investigation.	<code>/workorder-inbound-api/v1/work/notification</code>	POST

Note

Only inbound work order requests are supported using REST. You cannot send asynchronous responses through the REST interface.

JWT Authentication

The Work Order Inbound REST APIs are authenticated using the JWT based authentication, see [REST APIs for Accessing JWT](#) for details on how to obtain the JSON Web Token (JWT). The obtained JWT must be sent as a bearer token (HTTP authorization header) with every invocation of the REST API.

Application Error Codes

The following table describes the API response codes that can be returned for application errors.

Error Code	Error Message
2001	Invalid input data
2002	Server is unable to process the request
2011	Access denied

Common HTTP Codes

The following table lists the HTTP response codes the Work Order Inbound REST APIs can return.

HTTP Code	Model	Description
201	Success Response <pre> { "header": { "verb": "created", "noun": "work", "revision": "1", "timestamp": "2024-09-23T11:15:40", "source": "EnergyIP", "utility": "Sample", "messageID": "123", "property": [{ "name": "typeWork", "value": "OnDemandRead" }] }, "payload": { "genericWorkResponsePayload": { "wmsWorkId": { "id": "59" }, "externalWorkId": { "id": "ID-104", "idNamespace": "TEST" }, "statusCode": "In Progress", "reasonCode": "Done" } } } </pre>	Work Order Create REST API response which is an ack response for the DCT request.
400	Bad Request <pre> { "errorCode": "2001", "errorMessage": "Invalid input data [Name cannot be empty or null].", "httpResponseCode": 400 } </pre>	The request is not well-formed and the server does not understand it as it contains invalid values. The <code>errorMessage</code> and <code>errorCode</code> tags will be different as mentioned in the Application Error Codes.
403	Forbidden <pre> { "errorCode": 2011, "errorMessage": "Access denied", "httpResponseCode": 403 } </pre>	User does not have enough permissions.
401	Unauthorized	JWT is not present in the header as bearer token.
418	Invalid JWT	JWT present in the header is malformed or invalid.
419	JWT is expired	JWT present in the header is expired.

500	Internal Server Error <pre>{ "errorCode": 500, "errorMessage": "Some problem occurred while processing the request.", "httpResponseCode": 500 }</pre>	Something unusual happened at the server while processing the request.
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Create Work Order REST API

The Create Work Order REST API lets the external system to send a request to MDM to create a work order for a specified Device Control Transaction (DCT) operation.

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Supported Operation

Function	Operation	Method
Create a work order for a specified DCT operation	/workorder-inbound-api/v1/work/create	POST

(Open API documentation is only available to view online)

POST - Create Work Order

The POST method is used to create an inbound work order for the specified DCT request. This method requires the `agInterface.workRequest.create` permission to create work orders. For JWT authentication, see [Work Order Inbound REST APIs](#).

Input Parameters

The Create Work Order REST API accepts the `WorkRequestMessage` that contains header and payload sections.

Header

The following table describes the various fields in the header section of the request. These details include data type and descriptions of the input along with the fields that are relevant to the POST operation.

Parameter Name	Data Type	Format	Required	Description and Parameter Level Validations
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: create.
noun	String		Y	Identifies the main subject of the message type. Valid value: Work.
timestamp	DateTime	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Date and time when the message was collected.
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.
utility	String		Y	Identification for the utility, for example, the DUNS number of the utility.
replyAddress	String		N	The reply address for the Customer Information System (CIS). This address is used for sending responses or notifications back to the CIS. Essentially, it allows the MDM to know where to send any follow up messages or acknowledgments related to the work order request.

messageID	String		Y	Unique message Id used to track messages.
correlationID	String		N	Unique transaction Id used to correlate replies.
header:property - List of properties defining the name and value for the task (DCT operation). For example, name: typeWork and value: OnDemandRead				
name	String		Y	Name of the task. Valid value: typeWork
value	String		Y	Value of the task. The following values are supported: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration

Payload

The following table explains the payload section of the WorkRequestMessage. This table describes the fields common across all DCT operations, such as, On Demand Read, Set Load Limit, Check Load Limit, and so on. Certain DCT operations include additional, DCT operation specific fields, see [DCT Specific Request Payload Fields](#) for details.

Parameter Name	Format	Data Type	Required	Description
payload:genericWorkRequestPayload				
workPriority		String	N	Indicates the priority of the work order. Valid values: Numeric value from 1 to 4.
typeWork		String	Y	The work order type requested from the service provider. This work order type decides the service request/activity type and subtype. For all DCTs, the type of service request and activity is Meter Service. The subtype of service request and activity will be according to the type of the specified work order. The following work order types are supported. - RemoteCheckCDCStatus - The subtype of service request and activity will be Check Load Status. - OnDemandRead - The subtype of service request and activity will be On-Demand Read column of the Service_Request table. - PowerStatusVerification - The subtype of service request and activity will be Power-Verification column of the Service_Request table. - PowerStatusVerificationByGroup - The subtype of service request and activity will be Power-Verification. - HardConnect - The subtype of service request and activity will be Load - Hard Reconnect. - SoftConnect - The subtype of service request and activity will be Load - Soft Reconnect. - Disconnect - The subtype of service request and activity will be Load - Disconnect. - SendTextMessage - The subtype of service request and activity will be Send-Text Message. - CheckLoadLimit - The subtype of service request and activity will be Check Load Limit. - SetLoadLimit - The subtype of service request and activity will be Set Load Limit. - CheckMeterConfiguration - The subtype of service request and activity will be Check Meter Configuration. - SetMeterConfiguration - The subtype of service request and activity will be Set Meter Configuration.
remarks		String	N	Used for any freeform text or other text remarks. For example, check if meter is connected.
plannedStartDt	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	DateTime	N	Date on which work will be started on the work order. The service provider picks up request for execution on this date. If the planned start date is empty, the workflow process uses the system date.
plannedEndDt	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	DateTime	N	Date on which work needs to be end on the work order. If the planned end date is empty, the workflow process uses the system date. The planned end date must not before the planned start date.
payload:genericWorkRequestPayload:externalWorkId				
id		String	Y	The unique ID of the requester requesting service from the service provider.
idNamespace		String	N	Namespace of the request requesting service from the service provider.
payload:genericWorkRequestPayload:serviceDeliveryPoint				
id		String	Y	Unique Id of the service point.
idType		String	Y	Type of the service point Id. The following types are supported: - SDP_X_UNIVERSAL_ID - UdcSDPId
idNamespace		String	Y, if idType is UdcSDPId	Unique Id of the utility. For example, the DUNS number for the utility.
payload:genericWorkRequestPayload:meter				
id		String	Y	Unique Id of the meter. If service point Id is unavailable, the Activity Gateway will use the meter Id in a request. Activity Gateway requires a service point Id or meter Id.

idType		String	Y	Meter identification type. The following meter identification types are supported: - ElectronicMeterId - UdcMeterId - MeterBadgeld - Ansild
idNamespace		String	Y, if idType is Ansild or UdcMeterId	Unique value of the meter.

DCT Specific Request Payload Fields

The following subtopics describe the additional fields which can be added to **payload:genericWorkRequestPayload** based on the DCT operation.

On Demand Read

The following parameters are specific to the On Demand Read operation. These parameters are part of **payload:genericWorkRequestPayload:odrParams**.

Parameter Name	Data Type	Format	Mandatory	Description and Parameter Level Validations
startTime	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Read start time.
endTime	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Read end time.
measurementProfile	String		N	Used to fetch clean reads if available in the request.
demandResetMeasurementProfile	String		N	If the value of the demandResetRequired flag is set to Y, the demand reset measurement profile is passed as measurement profile to AMI, if it is present.
retrieveRegisterReads	Boolean		N	Flag to retrieve register reads.
retrieveIntervalReads	Boolean		N	Flag to retrieve interval reads.
retrieveEvents	Boolean		N	Flag to retrieve events.
meterReadCheckRequired	Boolean		N	Flag to perform meter read check.
demandResetRequired	Boolean		N	Flag to check for demand reset.

PowerStatusVerificationByGroup

The following parameters are specific to the multiple Power Status Verification (PSV) operation. These parameters are part of **payload:genericWorkRequestPayload:psvParams**.

Name	Data Type	Required	Description
payload:genericWorkRequestPayload:psvParams			
consolidatedResponse	Boolean	N	Flag to check if the streaming response or batch response is expected. See Power Status Verification By Group - Workflow Services for logic details.
reportCollectionStatusOnly	Boolean	N	Flag to check for response of distribution node only or for child svcs also. See Power Status Verification By Group - Workflow Services for logic details.
successThreshold	Integer	N	The minimum number of device responses required to consider a request as successful.
failureThreshold	Integer	N	The minimum number of device responses required to consider a request as failed.
payload:genericWorkRequestPayload:serviceDeliveryPointList			
serviceDeliveryPoint:id	String	Y	Service point identification value. The serviceDeliveryPoint Id specified in the payload must exist in the database and belong to the org of the logged in user.
serviceDeliveryPoint:idType	String	Y	Service point identification type. The following service point Id types are supported: - SDP_X_UNIVERSAL_ID - UdcSDPId
idNamespace	String	Y, if idType is UdcSDPId	Identification for the utility. For example, the DUNS number for the utility.

payload: genericWorkRequestPayload: meterList			
meter:id	String	Y	Meter identification value. If service point Id is unavailable, Activity Gateway will use the Meter Id in a request. The Activity Gateway requires either the service point Id or meter Id.
meter:idType	String	Y	Meter identification type. The following meter identification types are supported: • ElectronicMeterId • UdcMeterId • MeterBadgeld • Ansild
idNamespace	String	Y, if idType is Ansild or UdcMeterId	Unique value for the meter.
payload: genericWorkRequestPayload: distributionNode			
id	String	Y	Distribution Node identification value.
idType	String	Y	Distribution Node identification type. The supported distribution node identification type is UdcDistributionNodeId.

Connect/Disconnect

The following parameters are specific to Connect and Disconnect operations. These parameters are added as part of **payload:genericWorkRequestPayload:genericParams**.

Parameter Name	Data Type	Required	Description and Parameter Level Validations
parameter:name	String	Y	For a hard connect/disconnect work order request, populate the following generic parameter: • LoadSideVoltageCheckRequired For a hard connect/disconnect with an on-demand read, populate the following additional generic parameters: • MeterReadCheckRequired (optional) • RegisterDataRequired • IntervalDataRequired • EventDataRequired • DemandResetRequired • MeasurementProfile (optional) • DemandResetMeasProfile (optional) • CDCStatusCheck Required (optional) • ODR Required (optional) • ReadStartDateTime (optional) • ReadEndDateTime (optional)
parameter:value	String	Y	Value of the parameter. The possible values are true and false.

SendTextMessage

The following parameters are specific to Send Text Message operation. These parameters are added as part of **payload:genericWorkRequestPayload:textMsgParams**.

Parameter Name	Data Type	Format	Mandatory	Description and Parameter Level Validations	Default Value?
textMessage	String		Y	Text message which will be displayed on meter.	
messagePriority	String		N	Priority of the message.	
transmissionMode	String		Y	Transmission mode. Possible values: Anonymous or Normal. In case of anonymous, service request will not be created, and request will not be tracked.	
confirmationRequired	Boolean		N	Indicates whether confirmation is required or not.	
duration	Integer		N	Duration for which the text message will be displayed on meter. Valid values are zero or positive number.	
startDateTime	DateTime	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	Y	Time when message will start displaying.	
issuerId	String		N	Id of the issuer.	

issuerTrackingId	String		N	Tracking Id of the issuer.	
reason	String		N	Reason for displaying the message.	

CheckLoadLimit

The following parameters are specific to the Check Load Limit operation. These parameters are added as part of **payload:genericWorkRequestPayload:checkLoadLimitParams**.

Name	Data Type	Mandatory	Description and Parameter Level Validations
loadLimitProfile	String	N	Name of the load limit profile of the meter.
hasLoadLimitProfile	Boolean	N	Indicates if meter has a load limit profile.
issuerId	String	N	Id of the issuer.
issuerTrackingId	String	N	Tracking Id of the issuer.
reason	String	N	Reason for check load limit.
skipMdmCheck	Boolean	N	Specifies if the load limit profile needs to be validated against the MDM data. The possible values are true and false. False indicates that the load limit profile needs to be validated against the MDM data.

SetLoadLimit

The following parameters are specific to the Set Load Limit operation. These parameters are added as part of **payload:genericWorkRequestPayload:setLoadLimitParams**.

Name	Data Type	Format	Mandatory	Description and Parameter Level Validations
loadLimitProfile			N	Defines the fields of load limit profile as follows: - name: For example, SampleLoadLimitProfileName - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false. True indicates that load limit profile needs to be reset.
startDateTime	DateTime	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Time when load limit profile will be set.
emergencyProfileStartTime	DateTime	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Emergency profile start time.
emergencyProfileDuration	Integer		N	Duration for which emergency profile will be active. Valid values are zero or positive number.
issuerId	String		N	Id of the issuer.
issuerTrackingId	String		N	Tracking Id of the issuer.
reason	String		N	Reason for set load limit.
updateAssetInfo	Boolean		N	Specifies if asset needs to be updated in the MDM. True indicates asset will be updated in MDM.

CheckMeterConfiguration

The following parameters are specific to the Check Meter Configuration operation. These parameters are added as part of **payload:genericWorkRequestPayload:checkMeterConfigurationParams**.

Name	Data Type	Required	Description and Parameter Level Validations
programId	String	N	Id of the program which needs to run on a meter.
deriveMeasurementProfile	Boolean	Y	Specifies if the measurement profile needs to be driven from configParameter. The possible values are true and false. True indicates measurement profile needs to be driven from configParameter.
measurementProfile	String	N	Measurement profile name. This parameter is used if the value of the deriveMeasurementProfile parameter is false.

configParameter	Object	N	An array of parameter profiles. These parameters are used to provide billing and non-billing measurement profiles. The ParameterProfile defines the name, value, and clearValue fields. For example: - name: Billing Meas Profile - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false. - value: SampleBilling4
touProfile	Object	N	The Time of Unit (TOU) profile to be set on a meter. It defines the name and clearValue field as follows: - name: Sample TOU Profile Id - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false.
holidayProfile	Object	N	Holiday profile to be set on a meter. It defines the name and clearValue field as follows: - name: Sample Holiday Profile Id - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false.
loadControlTimeSwitchProfile	Object	N	Load control time switch profile to be set on a meter. It defines the name and clearValue field as follows: - name: Sample Load Control Time Switch Profile Id - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false.
loadLimitProfile	Object	N	Load limit profile to be set on a meter. It defines the name and clearValue field is as follows: - name: Sample Load Limit Profile Id - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false.
issuerId	String	N	Id of the issuer.
issuerTrackingId	String	N	Tracking Id of the issuer.
reason	String	N	Reason for which the check meter configuration operation needs to be performed.
skipMdmCheck	Boolean	N	Specifies if the load limit profile needs to be validated against the MDM data. The possible values are true and false. False indicates that the load limit profile needs to be validated against the MDM data.

SetMeterConfiguration

The following parameters are specific to the Set Meter Configuration operation. These parameters are added as part of **payload: genericWorkRequestPayload:setMeterConfigurationParams**.

Name	Data Type		Mandatory	Description and Parameter Level Validations
programId	String		N	Id of the program which needs to run on a meter.
deriveMeasurementProfile	Boolean		Y	Specifies if the measurement profile needs to be driven from configParameter. The possible values are true and false. True indicates measurement profile needs to be driven from configParameter.
measurementProfile	String		N	Measurement profile name. This parameter is used if the value of the deriveMeasurementProfile parameter is false.
configParameter	Object		N	An array of parameter profiles. These parameters are used to provide billing and non-billing measurement profiles. The ParameterProfile defines the name, value, and clearValue fields. For example: - name: Billing Meas Profile - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false. - value: SampleBilling4
touProfile	Object		N	The Time of Unit (TOU) profile to be set on a meter. It defines the name and clearValue field as follows: - name: Sample TOU Profile Id - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false.
holidayProfile	Object		N	Holiday profile to be set on a meter. It defines the name and clearValue field as follows: - name: Sample Holiday Profile Id - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false.

loadControlTimeSwitchProfile	Object		N	Load control time switch profile to be set on a meter. It defines the name and clearValue field as follows: - name: Sample Load Control Time Switch Profile Id - clearValue: Possible Indicates if the value present at the headend system needs to be cleared. The possible values are true and false.
loadLimitProfile	Object		N	Load limit profile to be set on a meter. It defines the name and clearValue field is as follows: - name: Sample Load Limit Profile Id - clearValue: Indicates if the value present at the headend system needs to be cleared. The possible values are true and false.
activationDate	Date	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Activation time of the profiles.
issuerId	String		N	Id of the issuer.
issuerTrackingId	String		N	Tracking Id of the issuer.
reason	String		N	Reason for which the set meter configuration operation needs to be performed.
skipMdmCheck	Boolean		N	Specifies if the load limit profile needs to be validated against the MDM data. The possible values are true and false. False indicates that the load limit profile needs to be validated against the MDM data.
updateAssetParameters	Boolean		N	Specifies if asset needs to be updated in the MDM. True indicates asset will be updated in MDM.
updateChannels	Boolean		N	Specifies if channels need to be updated in the MDM. True value indicates that channels need to be updated in MDM.
performFirstReadValidation	Boolean		N	Specifies if First Read Validation (FRV) needs to be performed on meter. True value indicates that FRV needs to be performed on meter.
doFinalRead	Boolean		N	Specifies if ODR operation needs to be performed before setting the meter configurations. True value indicates that ODR operation is performed before setting the meter configurations.
doInitialRead	Boolean		N	Specifies if ODR operation needs to be performed after setting the meter configurations. True value indicates that ODR operation is performed after setting the meter configurations.

Input Data Validations

The following validations are performed on the request payload.

Fields	Restrictions
header fields verb, noun, source, utility, messageId, property: name, property:value payload:genericWorkRequestPayload fields typeWork externalWorkId:id serviceDeliveryPoint:id, serviceDeliveryPoint: idType meter:id, meter:idType	- All the header and payload fields mentioned in this row are mandatory. Id and IdType of either serviceDeliveryPoint or meter is mandatory. - The valid value for verb field is create. - The serviceDeliveryPoint Id or meter Id specified in the payload must exist in the database and belong to the org of the logged in user. - The following values are supported for the property:value field in header and the typeWork field in payload: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
header:timestamp payload:genericWorkRequestPayload fields plannedStartDt plannedEndDt	The header and payload fields mentioned in this row should be specified in the ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format.

Response Details

The Create Work Order REST API returns an ack response that contains header and payload. The following table explains the header and payload sections.

Element	Data Type	Format	Required	Description
header				
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: created.
noun	String		Y	Identifies the main subject of the message type. Valid value: Work.
revision	String		N	Revision value Valid value: 1.
timestamp	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Date and time when the acknowledgment is sent.
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.
utility	String		Y	Identification for the utility, for example, the DUNS number for the utility.
messageID	String		Y	Unique message Id or transaction Id used to correlate replies and tracking messages.
correlationID	String		N	Unique transaction Id used to correlate replies.
property/name	String		Y	Defines the name and value for the task (DCT operation). For example, name: typeWork and value: OnDemandRead. The typeWork is the work order type requested from the service provider. This work order type decides the type and subtype of service request and activity. For all DCTs, the type of service request and activity is Meter Service. The subtype of service request and activity will be according to the type of the specified work order. The following values are supported for work order type: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
payload: genericWorkResponsePayload				
resultCode	String		Y	The derived code from status code.
statusCode	String		Y	The status of the request.
payload: genericWorkResponsePayload: externalWorkId				
id	String		Y	The unique Id of the requester requesting service from the service provider.
idNamespace	String		N	Namespace of the request requesting service from the service provider.
payload: genericWorkResponsePayload: wmsWorkId				
id	String		Y	The service request Id for which the ack response is provided.

POST - Create Work Order Request Example

In this example, an OnDemandRead request is sent for a service point to retrieve register reads and interval reads for a particular interval. The request will execute on the planned start date.

To create this work order, make a POST request to:

Request URL

`https://{host}:{port}/workorder-inbound-api/v1/work/create`

Use a payload of:

Sample Request

```
{
  "header": {
    "verb": "create",
    "noun": "work",
    "timestamp": "2024-09-04T03:50:00GMT",
    "utility": "Sample",
    "source": "Test",
    "messageID": "123",
    "correlationID": "CORRID1",
    "property": [
      {
        "name": "typeWork",
        "value": "OnDemandRead"
      }
    ]
  },
  "payload": {
    "genericWorkRequestPayload": {
      "externalWorkId": {
        "id": "ID-104",
        "idNamespace": "TEST"
      },
      "workPriority": "1",
      "typeWork": "OnDemandRead",
      "remarks": "CSR - Just Read",
      "plannedStartDt": "2024-09-12T12:30:00Z",
      "plannedEndDt": "2024-09-15T16:30:00Z",
      "odrParams": {
        "startTime": "2024-09-10T16:30:00Z",
        "endTime": "2024-09-12T16:30:00Z",
        "measurementProfile": "test_meas",
        "retrieveRegisterReads": true,
        "retrieveIntervalReads": true,
        "retrieveEvents": false,
        "meterReadCheckRequired": false,
        "demandResetRequired": false
      },
      "serviceDeliveryPoint": {
        "id": "SDP-UNI-10001",
        "idType": "SDP_X_UNIVERSAL_ID"
      }
    }
  }
}
```

The following code block shows the sample Create Work Order REST API sample ack response. The service request with Id 59 is created.

Sample Response

```
{
  "header": {
    "verb": "created",
    "noun": "work",
    "revision": "1",
    "timestamp": "2024-09-23T11:15:40",
    "source": "EnergyIP",
    "utility": "Sample",
    "messageID": "123",
    "property": [
      {
        "name": "typeWork",
        "value": "OnDemandRead"
      }
    ]
  },
  "payload": {
    "genericWorkResponsePayload": {
      "wmsWorkId": {
        "id": "59"
      },
      "externalWorkId": {
        "id": "ID-104",
        "idNamespace": "TEST"
      },
      "statusCode": "In Progress",
      "reasonCode": "Done"
    }
  }
}
```

Reschedule Work Order REST API

The Reschedule Work Order REST API allows an external system to request MDM to reschedule an existing work order for a specified Device Control Transaction (DCT) operation.

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Supported Operation

Function	Operation	Method
Reschedule an inbound work order for a specified DCT operation	/workorder-inbound-api/v1/work/reschedule	POST

POST - Reschedule Work Order

This POST method is used to reschedule an already created work order for the specified DCT request. This method requires the `agInterface.workRequest.reschedule` permission to reschedule work orders. For JWT authentication, see [Work Order Inbound REST APIs](#).

Input Parameters

The Reschedule Work Order REST API accepts the `WorkRescheduleRequestMessage` that contains header and payload sections. The following table describes the various fields in reschedule work order REST API request.

Parameter Name	Data Type	Format	Required	Description
header				
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: update
noun	String		Y	Identifies the main subject of the message type. Valid value: WorkSchedule
timestamp	DateTime	ISO 8601 format (yyy-MM-dd'T'HH:mm:ss)	N	Date and time when message was collected.
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.
utility	String		Y	Identification for the utility, for example, the DUNS number for the utility.
messageID	String		Y	Unique message Id or transaction ID used to correlate replies and tracking messages.
header:property - List of properties defining the name and value for the task (DCT operation). For example, name: typeWork and value: OnDemandRead				
name	String		Y	Name of the task. Valid value: typeWork

value	String		Y	Value of the task. The following values are supported: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
payload:genericRescheduleRequestPayload				
remarks	String		N	Used for any free form text or other text remarks.
reasonCode	String		N	Code of the reason due to which work order needs to be rescheduled.
plannedStartDt	DateTi me		N	Date on which work will be started on the work order.
plannedEndDt	DateTi me		N	Date on which work needs to be end on the work order.
option	String		N	To hold or release the hold of reschedule request. Valid values: Set_Hold, Release_Hold
payload:genericRescheduleRequestPay load:wmsWorkId				
id	String		Y	The service request Id for which the reschedule action is triggered.
payload:genericRescheduleRequestPay load:externalWorkId				
id	String		N	The unique Id of the requester requesting service from the service provider.
idNamespace	String		N	Namespace of the request requesting service from the service provider.

Input Data Validations

The following validations are performed on the request payload.

Fields	Restrictions
header fields verb, noun, source, utility, messageID, property: name, property:value payload:genericRescheduleRequestPayload fields wmsWorkId:id	- All the header and payload fields mentioned in this row are mandatory. - The valid value for verb field is update. - The valid value for noun field is workschedule. - The following values are supported for the property:value field in header and the typeWork field in payload: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
header:timestamp payload:genericRescheduleRequestPayload fields plannedStartDt plannedEndDt	- The header and payload fields mentioned in this row must be specified in the ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format. - The planned end date should not before the planned start date.

Response Details

The Reschedule Work Order REST API returns a response that contains header and payload. The following table describes the header and payload sections.

Element	Data Type	Format	Required	Description
header				
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: updated
noun	String		Y	Identifies the main subject of the message type. Valid value: workschedule
revision	String		N	Revision value. Valid value: 1
timestamp	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Date and time when the response was sent.
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.
utility	String		Y	Identification for the utility, for example, the DUNS number for the utility.
messageId	String		Y	Unique message Id or transaction Id used to correlate replies and tracking messages.
header:property - List of properties defining the name and value for the task (DCT operation). For example, name: typeWork and value: OnDemandRead				
name	String		Y	Name of the task. Valid value: typeWork
value	String		Y	Value of the task. The following values are supported: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
payload:genericWorkResponsePayload				
remarks	String		N	Used for any freeform text or other text remarks.
statusCode	String		Y	Status of the service request.
statusDate	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	Y	Time when the reschedule request is generated.
outcomeCode	String		N	Translated outcome code.
reasonCode	String		N	The status response reason code. This field specifies NA for a successful response. For a failed response, this field contains the failure reason.
payload:genericWorkResponsePayload:ad:wmsWorkId				
id	String		Y	The service request Id for which the response is provided.
payload:genericWorkResponsePayload:ad:externalWorkId				
id	String		N	The unique Id of the requester requesting service from the service provider.

POST - Reschedule Work Order Request Example

In this example, a reschedule request is sent for a service request with id 62. This will change the execution time of service request according to the planned start date and planned end time present in request.

To reschedule a work order, make a POST request to:

Request URL

`https://{host}:{port}/workorder-inbound-api/v1/work/reschedule`

Use a payload of:

Sample Request

```
{
  "header": {
    "verb": "update",
    "noun": "workschedule",
    "timestamp": "2024-09-23T03:50:00GMT",
    "utility": "Sample",
    "source": "Test",
    "messageID": "123",
    "property": [
      {
        "name": "typeWork",
        "value": "OnDemandRead"
      }
    ]
  },
  "payload": {
    "genericRescheduleRequestPayload": {
      "wmsWorkId": {
        "id": "62"
      },
      "externalWorkId": {
        "id": "ID-102",
        "idNamespace": "TEST"
      },
      "reasonCode": "Application Error",
      "remarks": "reschedule request",
      "plannedStartDt": "2025-01-27T11:33:47",
      "plannedEndDt": "2025-01-29T11:33:47"
    }
  }
}
```

The following code block shows the Reschedule Work Order REST API sample response. The service request with Id 62 is rescheduled.

Sample Response

```
{
  "header": {
    "verb": "updated",
    "noun": "workschedule",
    "revision": "1",
    "timestamp": "2024-09-23T11:15:40",
    "source": "EnergyIP",
    "utility": "Sample",
    "ackRequired": false,
    "messageID": "123",
    "property": [
      {
        "name": "typeWork",
        "value": "OnDemandRead"
      }
    ]
  },
  "payload": {
    "genericWorkResponsePayload": {
      "wmsWorkId": {
        "id": "62"
      },
      "externalWorkId": {
        "id": "ID-102"
      },
      "statusCode": "Open",
      "statusDate": "2025-01-20T11:51:17",
      "reasonCode": "Done",
      "outcomeCode": "R0:Rescheduled",
      "remarks": "Application Error"
    }
  }
}
```

Work Order Status REST API

The Work Order Status REST API allows an external system to fetch the status of a work order for a specified Device Control Transaction (DCT) operation.

- [Supported Operation](#)
- [POST - Check Work Order Status](#)
 - [Input Parameters](#)
 - [Input Data Validations](#)
 - [Response Details](#)
 - [POST - Work Order Status Request Example](#)

Supported Operation

Function	Operation	Method
Fetch the status of an inbound work order for a specified DCT operation	/workorder-inbound-api/v1/work/status	POST

POST - Check Work Order Status

This POST method is used to get the status of a work order for the specified DCT request. The external system sends a request to MDM to fetch the status of a work order. This method requires the `agInterface.workRequest.getStatus` permission to fetch the status of work orders. For JWT authentication, see [Work Order Inbound REST APIs](#).

Input Parameters

The Work Order Status REST API accepts the `WorkStatusRequestMessage` that contains header and payload sections. The following table describes the various fields in the Work Order Status REST API request.

Element	Data Type	Format	Required	Description
header				
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: get
noun	String		Y	Identifies the main subject of the message type. Valid value: WorkStatus
timestamp	DateTime	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Date and time when the message was collected.
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.
utility	String		Y	Identification for the utility, for example, the DUNS number for the utility.
messageID	String		Y	Unique message Id or transaction Id used to correlate replies and tracking messages.
header:property - List of properties defining the name and value for the task (DCT operation). For example, name: typeWork and value: OnDemandRead				
name	String		Y	Name of the task. Valid value: typeWork

value	String		Y	Value of the task. The following values are supported: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
payload:genericStatusRequestPayload				
payload:genericStatusRequestPayload:externalWorkId				
id	String		N	The unique Id of the requester requesting service from the service provider.
idNamespace	String		N	Namespace of the request requesting service from the service provider.
payload:genericStatusRequestPayload:wmsWorkId				
id	String		Y	The service request Id for which the status action is triggered.

Input Data Validations

The following validations are performed on the request.

Fields	Restrictions
header fields verb, noun, source, utility, messageId, property: name, property:value payload:genericStatusRequestPayload fields wmsWorkId:id	- All the header and payload fields mentioned in this row are mandatory. - The valid value for verb field is get. - The valid value for noun field is workStatus. - The following values are supported for the property:value field in header and the typeWork field in payload: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
header:timestamp	The timestamp field must be specified in the ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format.

Response Details

The Work Order Status REST API returns a response that contains header and payload. The following table describes the header and payload sections.

Element	Data Type	Format	Required	Description
header				
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: reply

noun	String		Y	Identifies the main subject of the message type. Valid value: workstatus
revision	String		N	Revision value. Valid value: 1
timestamp	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Date and time when the message was collected.
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.
utility	String		Y	Identification for the utility, for example, the DUNS number for the utility.
messageID	String		Y	Unique message Id or transaction Id used to correlate replies and tracking messages.
header:property - List of properties defining the name and value for the task (DCT operation). For example, name: typeWork and value: OnDemandRead				
name	String		Y	Name of the task. Valid value: typeWork
value	String		Y	Value of the task. The following values are supported: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
payload:genericWorkResponsePayload				
remarks	String		N	Used for any freeform text or other text remarks.
statusCode	String		Y	Status of the service request.
statusDate	Date Time		Y	Date on which the status of the service request is returned.
reasonCode	String		N	The status response reason code. This field specifies NA for a successful response. For a failed response, this field contains the failure reason.
plannedStartDt	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Date on which work will be started on the work order.
plannedEndDt	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Date on which work needs to be end on the work order.
payload:genericWorkResponsePayload:wmsWorkId				
id	String		Y	The service request Id for which the response is provided.
payload:genericWorkResponsePayload:externalWorkId				
id	String		N	The unique Id of the requester requesting service from the service provider.

POST - Work Order Status Request Example

In this example, a status request is sent for a service request with Id 62.

To fetch the status of a work order, make a POST request to:

Request URL

https://{host}:{port}/workorder-inbound-api/v1/work/status

Use a payload of:

Sample Request

```
{
  "header": {
    "verb": "get",
    "noun": "WorkStatus",
    "timestamp": "2024-09-23T03:50:00GMT",
    "utility": "Sample",
    "source": "Test",
    "messageID": "123",
    "property": [
      {
        "name": "typeWork",
        "value": "OnDemandRead"
      }
    ]
  },
  "payload": {
    "genericRescheduleRequestPayload": {
      "wmsWorkId": {
        "id": "62"
      },
      "externalWorkId": {
        "id": "ID-102",
        "idNamespace": "TEST"
      }
    }
  }
}
```

The following code block shows the status of a service request with Id 62.

Sample Response

```
{
  "header": {
    "verb": "reply",
    "noun": "workstatus",
    "revision": "1",
    "timestamp": "2025-01-23T11:15:40",
    "source": "EnergyIP",
    "utility": "Sample",
    "ackRequired": false,
    "messageID": "123",
    "property": [
      {
        "name": "typeWork",
        "value": "OnDemandRead"
      }
    ]
  },
  "payload": {
    "genericWorkResponsePayload": {
      "wmsWorkId": {
        "id": "62"
      },
      "externalWorkId": {
        "id": "ID-102",
        "idNamespace": "TEST"
      },
      "statusCode": "Open",
      "statusDate": "2025-01-23T11:51:17",
      "reasonCode": "Done",
      "remarks": "Application Error",
      "plannedStartDt": "2025-01-27T11:33:47",
      "plannedEndDt": "2025-01-29T11:33:47"
    }
  }
}
```


Cancel Work Order REST API

The Cancel Work Order REST API allows an external system to request MDM to cancel a work order for a specified Device Control Transaction (DCT) operation.

- [Supported Operation](#)
- [POST - Cancel Work Order](#)
 - [Input Parameters](#)
 - [Input Data Validations](#)
 - [Response Details](#)
 - [POST - Cancel Work Order Request Example](#)

Supported Operation

Function	Operation	Method
Cancel an inbound work order for a specified DCT operation	/workorder-inbound-api/v1/work/cancel	POST

POST - Cancel Work Order

This POST method is used to cancel an inbound work order for the specified DCT request. This method requires the `agInterface.workRequest.cancel` permission to cancel work orders. For JWT authentication, see [Work Order Inbound APIs](#).

Input Parameters

The Cancel Work Order REST API accepts the `WorkCancelRequestMessage` that contains header and payload sections. The following table describes the various fields in the work cancel request.

Element	Data Type	Format	Required	Description	Default Value?
header					
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: cancel.	
noun	String		Y	Identifies the main subject of the message type. Valid value: work.	
timestamp	DateTime	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Date and time when the message was collected.	
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.	
utility	String		Y	Identification for the utility, for example, the DUNS number for the utility.	
messageID	String		Y	Unique message Id or transaction Id used to correlate replies and tracking messages.	
header:property - List of properties defining the name and value for the task (DCT operation). For example, name: typeWork and value: OnDemandRead					
name	String		Y	Name of the task. Valid value: typeWork	
value	String		Y	Value of the task. The following valid values are supported: • RemoteCheckCDCStatus • OnDemandRead • PowerStatusVerification • PowerStatusVerificationByGroup • HardConnect • SoftConnect • Disconnect • SendTextMessage • CheckLoadLimit • SetLoadLimit • CheckMeterConfiguration • SetMeterConfiguration	

payload:genericCancelRequestPayload				
remarks	String		N	Used for any free form text or other text remarks.
reasonCode	String		N	Reason for cancelling the work order.
payload:genericCancelRequestPayload:wmsWorkId				
id	String		Y	The service request Id for which the cancel request is triggered.
payload:genericCancelRequestPayload:externalWorkId				
id	String		N	The unique Id of the requester requesting service from the service provider.
idNamespace	String		N	Namespace of the request requesting service from the service provider.

Input Data Validations

The following validations are performed on the request.

Fields	Restrictions
header fields verb, noun, source, utility, messageID, property: name, property:value payload:genericCancelRequestPayload fields wmsWorkId:id	- All the header and payload fields mentioned in this row are mandatory. - The valid value for verb field is cancel. - The valid value for noun field is work. - The following values are supported for the property:value field in header and the typeWork field in payload: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
header:timestamp	The timestamp field must be specified in the ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format.

Response Details

The Cancel Work Order REST API returns a response that contains header and payload. The following table describes the header and payload sections.

Element	Data Type	Format	Mandatory	Description
header				
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: canceled.
noun	String		Y	Identifies the main subject of the message type. Valid value: work.
revision	String		N	Revision value. Valid value: 1
timestamp	Date Time	ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format	N	Date and time when the message was collected.
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.

utility	String		Y	Identification for the utility, for example, the DUNS number for the utility.
messageId	String		Y	Unique message Id or transaction Id used to correlate replies and tracking messages.
header:property - List of properties defining the name and value for the task (DCT operation). For example, name: typeWork and value: OnDemandRead				
name	String		Y	Name of the task. Valid value: typeWork
value	String		Y	Value of the task. The following values are supported: - RemoteCheckCDCStatus - OnDemandRead - PowerStatusVerification - PowerStatusVerificationByGroup - HardConnect - SoftConnect - Disconnect - SendTextMessage - CheckLoadLimit - SetLoadLimit - CheckMeterConfiguration - SetMeterConfiguration
payload:genericWorkResponsePayload				
remarks	String		N	Used for any freeform text or other text remarks.
statusCode	String		Y	Status of the service request.
outcomeCode	String		N	Translated outcome code.
reasonCode	String		N	The status response reason code. This field specifies NA for a successful response. For a failed response, this field contains the failure reason.
payload:genericWorkResponsePayload:ad:wmsWorkId				
id	String		Y	The service request Id for which the response is provided.
payload:genericWorkResponsePayload:ad:externalWorkId				
id	String		N	The unique Id of the requester requesting service from the service provider.

POST - Cancel Work Order Request Example

In this example, a request is sent to cancel the work order associated with the service request Id 59.

To create this cancel work order request, make a POST request to:

Request URL

`https://{host}:{port}/workorder-inbound-api/v1/work/cancel`

Use a payload of:

Sample Request

```
{
  "header": {
    "verb": "cancel",
    "noun": "work",
    "timestamp": "2024-09-23T03:50:00GMT",
    "utility": "Sample",
    "source": "Test",
    "messageID": "123",
    "property": [
      {
        "name": "typeWork",
        "value": "OnDemandRead"
      }
    ]
  },
  "payload": {
    "genericCancelRequestPayload": {
      "externalWorkId": {
        "id": "59"
      },
      "remarks": "Cancel the work order associated with the service request Id 59."
    }
  }
}
```

The following code block shows the sample Cancel Work Order REST API sample ack response. The service request with Id 59 is cancelled.

Sample Response

```
{
  "header": {
    "verb": "canceled",
    "noun": "work",
    "revision": "1",
    "timestamp": "2024-09-23T11:15:40",
    "source": "EnergyIP",
    "utility": "Sample",
    "ackRequired": false,
    "messageID": "123",
    "property": [
      {
        "name": "typeWork",
        "value": "OnDemandRead"
      }
    ]
  },
  "payload": {
    "genericWorkResponsePayload": {
      "wmsWorkId": {
        "id": "59"
      },
      "externalWorkId": {
        "id": "ID-101",
        "idNamespace": "TEST"
      },
      "statusCode": "Done",
      "outcome": "R0:Cancelled"
    }
  }
}
```

Work Status Notification REST API

The Work Status Notification REST API allows an external system to send a work status notification to MDM when a field investigation is done or when updates are made to the field investigation. It helps MDM know about the complete history of the work performed at the meter in the field.

- [Supported Operation](#)
- [POST - Work Status Notification](#)
 - [Input Parameters](#)
 - [Request Parameter](#)
 - [Header](#)
 - [Payload](#)
 - [Attachment Parameter](#)
 - [Input Data Validations](#)
 - [Response Details](#)
 - [WorkType Attribute Maps](#)
 - [POST - Work Status Notification Request Example](#)

Supported Operation

Function	Operation	Method
Send a work status notification to MDM when a field investigation is done or when updates are made to the field investigation.	/workorder-inbound-api/v1/work/notification	POST

POST - Work Status Notification

This POST method is used to send the work status notification to MDM when a field investigation is done or when updates are made to the field investigation. This method requires the `agInterface.workRequest.replicate` permission to send the work status notification message. For JWT authentication, see [Work Order Inbound REST APIs](#).

Input Parameters

The Work Order Notification REST API is used to initiate a work status notification request. You need to send a request and an image type attachment as parameters with the `multipart/form-data` encoding type. The `multipart/form-data` encoding type allows request and images (evidence of work performed at meter) to be sent to the REST API endpoint through a POST method. Both the request and attachment parameters are mandatory. However, the attachment parameter can be null if you do not want to upload any evidence.

Request Parameter

The request Parameter is of `WorkStatusNotificationRequest` type which consist of header and payload sections. The `ContentType` of the request must be `application/json`. See [POST - Work Status Notification Request Example](#) for a sample request.

Header

Name	Data Type	Format	Mandatory	Description
header				
noun	String		Y	Identifies the main subject of the message type. Valid value: WorkNotification.
verb	String		Y	Identifies the action of the source. Verb is case sensitive. Valid value: update.
timestamp	Date Time	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Date and time when the message was generated.
source	String		Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.
utility	String		Y	Identification for the utility. For example, the DUNS number for the utility.
messageID	String		Y	Unique message Id or transaction Id used to correlate replies and tracking messages.

- **header:property** - Properties associated with a message that help determine how the message should be processed, routed, and filtered. For example, where the message should go and which messages should be processed under certain conditions. The message property consists of two components:
 - **Name:** Name of the task. For example, typeWork.
 - **Value:** Defines the specific type of work being referred to. For example, ReBillCustomer-RegularRead, where ReBillCustomer relates to a customer rebilling process and regular read refer to a regular meter reading.

name	String		Y	Name of the task. Valid value: typeWork
value	String		Y	Type of external work. You can create any type of external work, for example, • ReBillCustomer-RegularRead • ReBillCustomer-SORead • MeterInvestigation-ExcessiveMissingReads • MeterInvestigation-MissingReads • UsageOnDisconnMeter • UsageOnVirtualDisconnMeter • UsageonDisconnService • Diversion-TheftFlags • Diversion-UnauthorizedUsage • MI (Meter Investigation)

Payload

The following table explains the payload section of the WorkStatusNotificationRequest.

Parameter Name	Data Type	Format	Mandatory	Description
payload:genericWorkStatusNotificationPayload				
workPriority	String		N	Indicates work order priority. Valid values: Numeric value from 1 to 4.
typeWork	String		N	The external work type requested from the service provider. For the main task (that maps to a service request), the value of typeWork must be equal to the property value present in the header. However, for subtasks (activities), the value of typeWork can be different from the property value of the header. Some of the supported external work types are listed below: • ReBillCustomer-RegularRead • ReBillCustomer-SORead • MeterInvestigation-ExcessiveMissingReads • MeterInvestigation-MissingReads • UsageOnDisconnMeter • UsageOnVirtualDisconnMeter • UsageonDisconnService • Diversion-TheftFlags • Diversion-UnauthorizedUsage • MI (Meter Investigation) These external work order type decides the type and subtype of service request and activity. See WorkType Attribute Maps for information on the type and subtype of service request and activity.
statusCode	String		N	Indicates the status of the work order. If a status code is not specified, the workorder-inbound-api application sets the status to Open. The following values are supported. • Open • Done • Failed • Pending • Cancelled • Hold
statusDate	DateTime	ISO 8601 format (yyyy-MM-ddTHH:mm:ss)	N	Used only when the work order status is Open or Done. • If the status is Open, this field displays the value of the Date Opened field of the service request in the UI. • If the status is Done, this field displays the Date Closed field of the service request. The workorder-inbound-api application ignores the status date for all other status codes or if the status code field is blank.
outcomeCode	String		N	Outcome code.
remarks	String		N	Used for any freeform text or other text remarks.
payload:genericWorkStatusNotificationPayload:wmsWorkId				
id	String		Y	Unique Id generated by the external system for the work order.
payload:genericWorkStatusNotificationPayload:externalWorkId				
id	String		Y	The unique Id of the requester requesting service from the service provider.
idNamespace	String		N	Namespace of the request requesting service from the service provider.
payload:genericWorkStatusNotificationPayload:workTasks - The workorder-inbound-api application uses this element to create the associated activities for the service request. Each work task represents an activity in Gridscale X Meter Data Management.				
workTaskId:id	String		Y	Id generated by the service provider as part of external subtask (activity).

typeWorkTask	String		Y	Type of work for the subtask. Some of the supported subtasks are listed below: • ReBillCustomer-RegularRead • ReBillCustomer-SORead • MeterInvestigation-ExcessiveMissingReads • MeterInvestigation-MissingReads • UsageOnDisconnMeter • UsageOnVirtualDisconnMeter • UsageonDisconnService • Diversion-TheftFlags • Diversion-UnauthorizedUsage • MI (Meter Investigation)
taskPriority	String		N	Priority of the subtask.
plannedStartDt	DateTime	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Date and time when the work order needs to be started by service provider.
plannedEndDt	DateTime	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Date and time when the request needs to be completed by service provider.
actualStartDt	DateTime	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Date and time when the work order is started by service provider.
actualEndDt	DateTime	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Date and time when the work order is actually completed by service provider.
statusCode	String		N	Status of the subtask of the work order. The following values are supported: • Open • Done • Failed • Pending • Cancelled • Hold If a status code is not specified, the workorder-inbound-api application sets the status to Open.
statusDate	DateTime	ISO 8601 format (yyyy-MM-dd'T'HH:mm:ss)	N	Used only when the status of the subtask is cancelled or failed. The workorder-inbound-api application ignores the status code for all other status codes. This status date will be displayed in the Failed or canceled field of the activity in the UI.
remarks	String		N	Freeform text or canned text remarks.
reasonCode	String		N	Reason code.
payload:genericWorkStatusNotificationPayload:serviceDeliveryPoint				
id	String		Y	Unique Id of the service point. If the service point Id is not available, the workorder-inbound-api application uses the Meter Id. The workorder-inbound-api application either requires a service point Id or meter Id.
idType	String		Y	Type of the service point Id. The following types are supported: • SDP_X_UNIVERSAL_ID • UdcSDPId
idNamespace	String		Y, if idType is UdcSDPId	Unique Id of the utility. For example, the DUNS number for the utility.
payload:genericWorkStatusNotificationPayload:meter				
id	String		Y	Unique Id of the meter. If the service point Id is not available, the workorder-inbound-api application uses the Meter Id. The workorder-inbound-api application either requires a service point Id or meter Id.
idType	String		Y	Meter identification type. The following meter identification types are supported: • ElectronicMeterId • UdcMeterId • MeterBadgeld • Ansild
idNamespace	String		Y, if idType is Ansild or UdcMeterId	Unique value of the meter.

Attachment Parameter

You can upload an evidence of the investigation performed at field in the attachment parameter. The evidence is an image type file. You can upload multiple images in a request. The supported image file types are .jpeg, .jpg, .png, .bmp, .gif, and .bm. See [Work Status Notification](#) for details.

The attachment parameter is mandatory. However, the attachment parameter can be null if you do not want to upload any evidence.

Input Data Validations

The Work Status Notification API accepts the request in the the multipart/form-data encoding typ. You have to provide the request and attachment parameters. The following validations are performed on the request payload.

Parameter	Fields	Restrictions
Request JSON format for WorkStatusNotificationRequest	WorkStatusNotificationRequest Request Sections header fields verb, noun, source, utility, messageId, property:name, property:value payload: genericWorkStatusNotificationPayload Fields wmsWorkId:id externalWorkId:id workTasks:workTaskId:id workTasks:typeWorkTask serviceDeliveryPoint:id, serviceDeliveryPoint:idType meter:id, meter:idType	- All the header and payload fields mentioned in this row are mandatory. Id and IdType of either serviceDeliveryPoint or meter is mandatory. - The valid value for verb field is update. - The serviceDeliveryPoint Id or meter Id specified in the payload must exist in the database and belong to the org of the logged in user.
	header:timestamp payload: genericWorkStatusNotificationPayload fields statusDate workTasks:plannedStartDt workTasks:plannedEndDt workTasks:actualStartDt workTasks:actualEndDt workTasks:statusDate	The header and payload fields mentioned in this row should be specified in the ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format.
Attachment		The valid image types are .jpeg, .jpg, .png, .bmp, .gif, and .bm.

Response Details

The following table describes the work response elements of a Work Status Notification REST API.

Name	Data Type	Mandatory	Description
header			
noun	String	Y	Identifies the main subject of the message type. Valid value: WorkNotification.
verb	String	Y	Identifies the action of the source. Verb is case sensitive. Valid value: updated.
revision	String	N	Revision level of message type. Valid value: 1.
timestamp	DateTime	N	Date and time when the message was produced. Specified in the ISO 8601 (yyyy-MM-dd'T'HH:mm:ss) format.
source	String	Y	Source person or system that published the message. For example, the DUNS number that identifies the work management system or the utility that generated the request.
correlationID	String	Y	Unique message Id or transaction Id used to correlate the response to request.

header:property - Properties associated with a message that help determine how the message should be processed, routed, and filtered. For example, where the message should go and which messages should be processed under certain conditions. The message property consists of two components:

- **Name:** Name of the task. For example, typeWork.
- **Value:** Defines the specific type of work being referred to. For example, ReBillCustomer-RegularRead, where ReBillCustomer relates to a customer rebilling process and regular read refer to a regular meter reading.

name	String	Y	Name of the task. Valid value: typeWork
value	String	Y	Type of external work. You can create any type of external work, for example, • ReBillCustomer-RegularRead • ReBillCustomer-SORead • MeterInvestigation-ExcessiveMissingReads • MeterInvestigation-MissingReads • UsageOnDisconnMeter • UsageOnVirtualDisconnMeter • UsageonDisconnService • Diversion-TheftFlags • Diversion-UnauthorizedUsage • MI (Meter Investigation)
payload:genericWorkResponsePayload			
wmsWorkId:id	String	Y	The main task Id.
outcomeCode	String	Y	Indicates success or failure.
reasonCode	String	Y (only in case of error or exception response)	Indicates reason for failure.
statusCode	String	N	Status of the service request.
resultCode	String	N	Retrieved from outcome and status code of the service request.

WorkType Attribute Maps

The Work Status Notification REST API maps the external work type to the type of service request/activity of Gridscale X Meter Data Management using value maps. Gridscale X Meter Data Management supports the field service work orders mentioned in the External Work Type column in the following table. The workorder-inbound-api creates or updates the service request according to the values in the Work Status Notification request. The service request type, service request subtype, activity type, and activity subtype will vary depending on the type of the external work request.

External Work Type	SR Type / Subtype	Activity Type / Subtype
ReBillCustomer-RegularRead	Billing – Estimated / ReBill Customer	WMS Field Activity / Regular Read Received
ReBillCustomer-SORead	Billing – Estimated / ReBill Customer	WMS Field Activity / SO Read Received
MeterInvestigation-ExcessiveMissingReads	PIPe Report / Excessive Missing Reads	WMS Field Activity / Meter - Investigation
MeterInvestigation-MissingReads	PIPe Report / Missing Reads	WMS Field Activity / Meter - Investigation
UsageOnDisconnMeter	PIPe Report / Out Of Sync	WMS Field Activity / Usage on Disconn Meter
UsageOnVirtualDisconnMeter	PIPe Report / Out Of Sync	WMS Field Activity / Usage on Virt Disconn Meter
UsageonDisconnService	PIPe Report / Out Of Sync	WMS Field Activity / Usage on Disconn Service
Diversion-TheftFlags	PIPe Report / Theft Flags	WMS Field Activity / Diversion
Diversion-UnauthorizedUsage	PIPe Report / Unauthorized Usage	WMS Field Activity / Diversion
MI	PIPe Report / Validation Error	WMS Field Activity / Meter - Investigation

Note

The value of the External Work Type column in the above table can be customized. This means a utility can have another work type to represent a diversion or unauthorized usage. However, the type and subtype of service request and activity cannot be customized. The external diversion or unauthorized usage work type must always be mapped to the service request and activity information mentioned in the table.

POST - Work Status Notification Request Example

In this example, a Work Status Notification request is sent for a service point whose universal id is "UWSN1001".

To send a work status notification, make a POST request to:

Request URL

https://{host}:{port}/workorder-inbound-api/v1/work/notification

Use the following sample JSON in the request parameter and provide evidence in the attachment parameter of multipart/form-data encoding type request.

Sample Request

```
{
  "header": {
    "verb": "update",
    "noun": "WorkNotification",
    "timestamp": "2001-12-31T12:00:00Z",
    "source": "DCTORG",
    "utility": "DCTORG",
    "messageID": "Txnid-WSN1001",
    "property": [
      {
        "name": "typeWork",
        "value": "FieldServiceWorkType"
      }
    ]
  },
  "payload": {
    "genericWorkStatusNotificationPayload": {
      "wmsWorkId": {
        "id": "Txnid-WSN1001",
        "idNamespace": "DCTORG"
      },
      "workPriority": "1",
      "typeWork": "FieldServiceWorkType",
      "statusCode": "Done",
      "outcomeCode": "46",
      "statusDate": "2009-03-31T12:00:00",
      "remarks": "main task",
      "workTasks": [
        {
          "workTaskId": {
            "id": "Txnid-WSN100101"
          },
          "typeWorkTask": "FieldServiceWorkType",
          "taskPriority": "1",
          "plannedStartDt": "2022-12-31T12:00:00",
          "plannedEndDt": "2023-12-31T12:00:00",
          "actualStartDt": "2023-12-31T12:00:00",
          "actualEndDt": "2024-12-31T12:00:00",
          "statusCode": "Done",
          "statusDate": "2024-12-31T12:00:00",
          "remarks": "sub task 1"
        },
        {
          "workTaskId": {
            "id": "Txnid-WSN100102"
          },
          "typeWorkTask": "FieldServiceWorkType2",
          "taskPriority": "1",
          "plannedStartDt": "2022-12-31T12:00:00",
          "plannedEndDt": "2023-12-31T12:00:00",
          "actualStartDt": "2023-12-31T12:00:00",
          "actualEndDt": "2024-12-31T12:00:00",
          "statusCode": "Done",
          "statusDate": "2024-12-31T12:00:00",
          "remarks": "sub task 2"
        }
      ],
      "workTaskId": {

```

```

        "id": "Txnid-WSN100103"
      },
      "typeWorkTask": "FieldServiceWorkType3",
      "taskPriority": "1",
      "plannedStartDt": "2022-12-31T12:00:00",
      "plannedEndDt": "2023-12-31T12:00:00",
      "actualStartDt": "2022-12-31T12:00:00",
      "actualEndDt": "2024-12-31T12:00:00",
      "statusCode": "Done",
      "statusDate": "2024-12-31T12:00:00",
      "remarks": "sub task 3"
    }
  ],
  "serviceDeliveryPoint": {
    "id": "UWSN1001",
    "idType": "SDP_X_UNIVERSAL_ID"
  }
}
}
}
}

```

The following code block shows the sample response from MDM to external system about the work status notification request.

Sample Response

```

{
  "header": {
    "verb": "updated",
    "noun": "WorkNotification",
    "revision": "1",
    "timestamp": "2024-09-23T11:15:40",
    "source": "EnergyIP",
    "utility": "Sample",
    "correlationID": "TRANSACTIONID",
    "property": [
      {
        "name": "typeWork",
        "value": "ReBillCustomer-RegularRead"
      }
    ]
  },
  "payload": {
    "genericWorkResponsePayload": {
      "wmsWorkId": {
        "id": "62"
      },
      "statusCode": "Done",
      "reasonCode": "Done",
      "outcomeCode": "FS:Successful"
    }
  }
}

```